

- 0: No delay.
- 1: Add 50ns delay. (Default value).

SDA_OUTPUT_DELAYED

SDA_OUTPUT_DELAYED attribute is used to add 50ns additional delay to the SDAO signal.

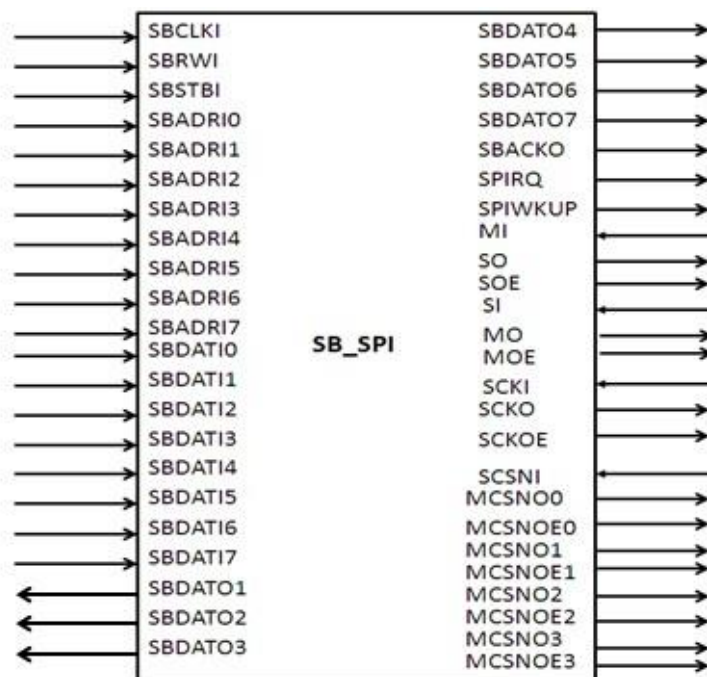
```
/* synthesis SDA_OUTPUT_DELAYED= value */
```

Value:

- 0: No delay (Default value).
- 1: Add 50ns delay.

SB_SPI

iCE5LP device supports two SPI hard IP primitives, located at lower left corner and lower right corner of the chip. The port interface of SB_SPI is similar as ICE40LM SB_SPI primitive. Refer Page 117.



Ports

The port interface is similar as iCE40LM SB_SPI primitive. Refer Page 117.

Parameters

The parameters are same as ICE40LM SB_SPI primitive.

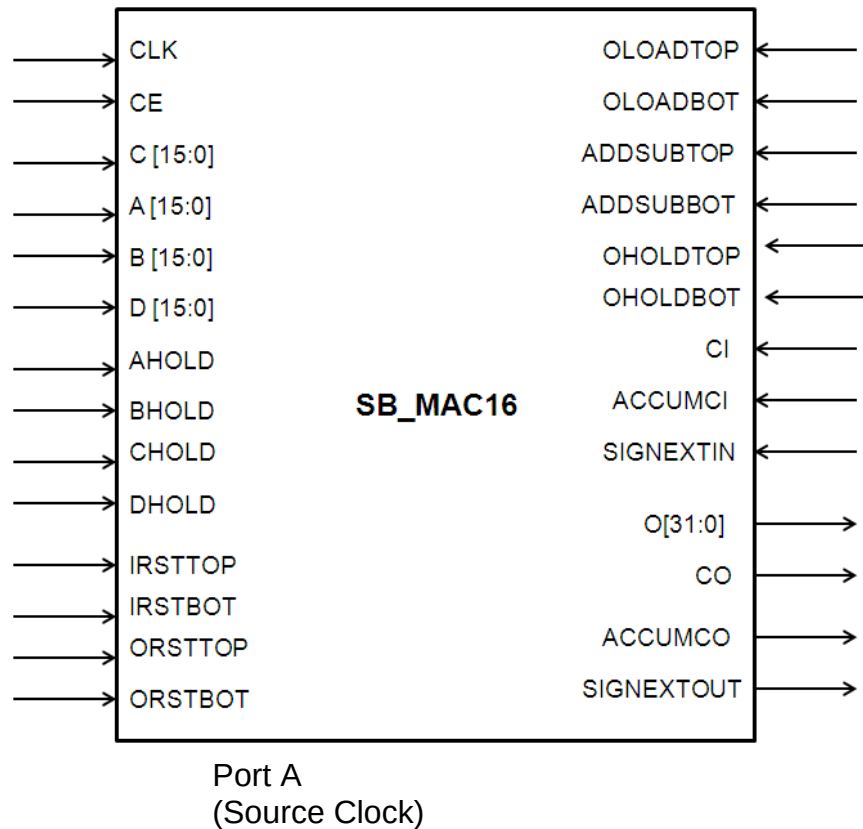
Synthesis Attribute

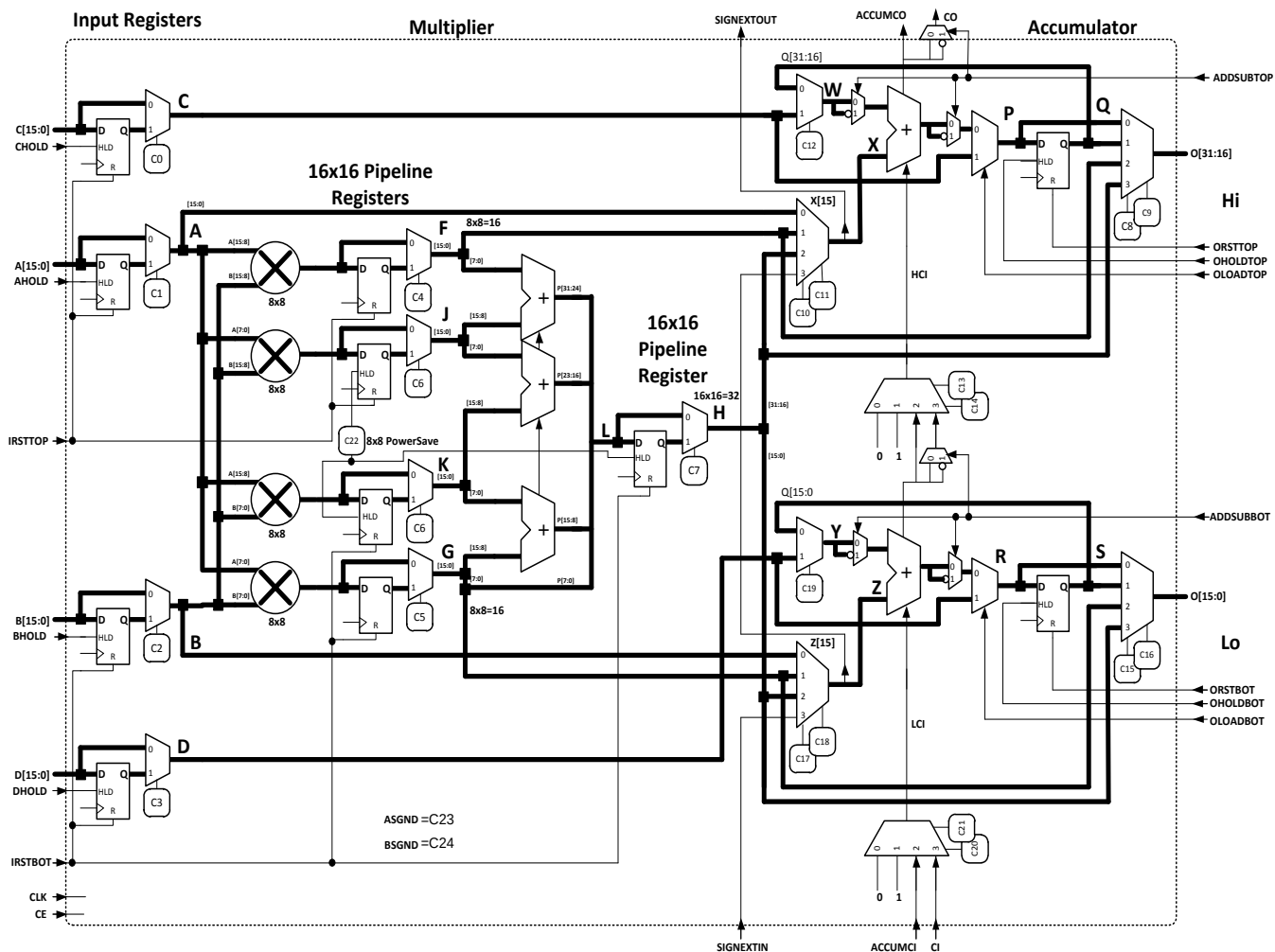
The synthesis attribute is same as ICE40LM SB_SPI primitive.

SB_MAC16

The SB_MAC16 primitive is the dedicated configurable DSP block available in iCE5LP devices. The SB_MAC16 can be configured into a multiplier, adder, subtracter, accumulator, multiply-add and multiply-sub through the instance parameters. The SB_MAC16 blocks can be cascaded to implement wider functional units.

iCEcube2 supports a set of predefined SB_MAC16 functional configurations. Refer Page 138 for the list of supported configurations.





Port A (Generated Clock)

Ports

SB_MAC16 Ports			
Signal Name	Direction	Description	Default Values
CLK	Input	Clock input. Applies to all clocked elements in the MAC16 Block.	
CE	Input	Clock Enable Input. Active High.	1'b1
C[15:0]	Input	Input to the C Register / Direct input to the adder accumulator.	16'b0
A[15:0]	Input	Input to the A Register / Direct input to the multiplier blocks /Direct input to the adder accumulator.	16'b0
B[15:0]	Input	Input to the B Register / Direct input to the multiplier blocks /Direct input to the adder accumulator.	16'b0
D[15:0]	Input	Input to the D Register / Direct input to the adder accumulator.	16'b0
AHOLD	Input	Hold A registers Data .Controls data flow into the input register A. Active High.	1'b0
		0 Update (load) register at next active clock edge.	

		1	Hold (retain) current register value, regardless of clock.	
BHOLD	Input	Hold B registers Data .Controls data flow into the B input register. Active High.		1'b0
		0	Update (load) register at next active clock edge.	
		1	Hold (retain) current register value, regardless of clock.	
CHOLD	Input	Hold C registers Data. Controls data flow into the A input register. Active High.		1'b0
		0	Update (load) register at next active clock edge.	
		1	Hold (retain) current register value, regardless of clock.	
DHOLD	Input	Hold D registers Data. Controls data flow into the A input register. Active High.		1'b0
		0	Update (load) register at next active clock edge.	
		1	Hold (retain) current register value, regardless of clock.	
IRSTTOP	Input	Reset input to the A and C input registers, and the pipeline registers in the upper half of the multiplier block. Active High		1'b0
IRSTBOT	Input	Reset input to the B and C input registers, and the pipeline registers in the lower half of the multiplier block, and the 32-bit multiplier result pipeline register. Active High.		1'b0
ORSTTOP	Input	Reset the high-order bits of the accumulator register ([31:16]). Active High.		1'b0
ORSTBOT	Input	Reset the low-order accumulator register bits ([15:0]). Active High.		1'b0
OLOADTOP	Input	High-order Accumulator Register Accumulate/Load. Controls whether the accumulator register accepts the output of the adder/subtractor or whether the register is loaded with the value from Input C (or Register C, if configured).		1'b0
		0	Accumulator Register [31:16] loaded with output from adder/subtractor.	
		1	Accumulator Register [31:16] loaded with Input C or Register C, depending on primitive parameter value.	
OLOADBOT	Input	Low-order Accumulator Register Accumulate/Load. Controls whether the low-order accumulator register bits (15:0] accepts the output of the adder/subtractor or whether the register is loaded with the value from Input D (or Register D, if configured).		1'b0
		0	Accumulator Register [15:0] loaded with output from adder/subtractor.	
		1	Accumulator Register [15:0] loaded with Input D or Register D, depending on primitive parameter value.	
ADDSUBTOP	Input	High-order Add/Subtract. Controls whether the adder/subtractor adds or subtracts.		1'b0: Add
		0	Add: W+X+HCI	
		1	Subtract: W-X-HCI	
ADDSUBBOT	input	Low-order Add/Subtract. Controls whether the adder/subtractor adds or subtracts.		1'b0: Add

		0	Add: Y+Z+LCI	
		1	Subtract: Y-Z-LCI	
OHOLDTOP	Input	High-order Accumulator Register Hold. Controls data flow into the high-order ([31:16]) bits of the accumulator.		
		0	Update (load) register at next active clock edge.	
		1	Hold (retain) current register value, regardless of clock.	
OHOLDBOT	Input	Low-order Accumulator Register Hold. Controls data flow into the high-order ([15:0]) bits of the accumulator.		
		0	Update (load) register at next active clock edge.	
		1	Hold (retain) current register value, regardless of clock.	
CI	Input	Carry/borrow input from lower logic tile.		
ACCUMCI	Input	Cascade Carry/borrow input from lower MAC16 block.		
SIGNEXTIN	Input	Sign extension input from lower MAC16 block.		
O[31:0]	Output	32 bit MAC16 Output.		
		O[31:0]	32-bit result of a 16x16 multiply operation or a 32-bit adder/accumulate function.	
		O[31:16]	16-bit result of an 8x8 multiply operation or a 32-bit adder/accumulate function.	
		O[15:0]	16-bit result of an 8x8 multiply operation or a 32-bit adder/accumulate function.	
CO	Output	Carry/borrow output to higher logic tile.		
ACCUMCO	Output	Cascade Carry/borrow output to higher MAC16 block.		
SIGNEXTOUT	Output	Sign extension output to higher MAC16 block.		

Parameters

The parameter table below shows the list of parameters to configure the SB_MAC16 block. This table also maps the parameter to the configuration bits shown in the SB_MAC16 Functional diagram.

Parameter Name	Configuration bits	Parameter Description	Parameter Values	Description
NEG_TRIGGER		Controls input clock polarity	0	All the registers are rising_edge triggered.
			1	All the registers are falling_edge triggered.
C_REG	C0	Input C register Control.	0	Input C not registered
			1	Input C registered
A_REG	C1	Input A register Control	0	Input A not registered
			1	Input A registered
B_REG	C2	Input B register Control	0	Input B not registered
			1	Input B registered
D_REG	C3	Input D register Control	0	Input D not registered
			1	Input D registered
TOP_8x8_MULT_REG	C4	Top 8x8 multiplier output register control (point F)	0	Top 8x8 multiplier output is not registered.
			1	Top 8x8 multiplier output is registered.
BOT_8x8_MULT_REG	C5	Bottom 8x8 multiplier	0	Bottom 8x8 multiplier

		output register control. (point G)		output is not registered.
			1	Bottom 8x8 multiplier output is registered
PIPELINE_16x16_MULT_REG1	C6	Intermediate 8x8 multiplier pipeline register controls (points J and K). These multipliers are only used for 16x16 multiply operations. For 8x8 multiply operations, set C6 and C7 to 1, which reduces power consumption.	0	Intermediate 8x8 multiplier outputs are not registered.
			1	Intermediate 8x8 multiplier outputs are registered.
PIPELINE_16x16_MULT_REG2	C7	16x16 multiplier Pipeline registers control. (point H)	0	32-bit output from 16x16 multiplier is not registered.
			1	32-bit output from 16x16 multiplier is registered.
TOPOUTPUT_SELECT	C9,C8	Selects Top SB_MAC16 output O[31:16].	00	16-bit output of Multiplexer P, from top adder/subtractor,
			01	16-bit output from upper accumulator register, Q
			10	16-bit output from upper 8x8 multiplier, F
			11	Upper 16-bit output from 16x16 multiplier, H
TOPADDSUB_LOWERINPUT	C11,C10	Selects input X for the upper adder/subtractor	00	16-bit input from A input or associated input register.
			01	16-bit output from upper 8x8 multiplier, F
			10	Upper 16-bit output from 16x16 multiplier, H
			11	Sign extension input from lower adder/subtractor, Z[15]. Duplicate 16 bits.
TOPADDSUB_UPPERINPUT	C12	Selects input W for the upper adder/subtractor	0	16-bit feedback from upper accumulator register, Q
			1	16-bit input from C input or associated pipeline register
TOPADDSUB_CARRYSELECT	C14,C13	Carry/borrow input select to upper adder/subtractor.	00	Constant 0
			01	Constant 1
			10	Cascade Carry/borrow output from lower adder/subtractor
			11	Carry/borrow output from lower adder/subtractor
BOTOUTPUT_SELECT	C16,C15	Selects Lower SB_MAC16 output O[15:0].	00	16-bit output of multiplexer R from lower adder/subtractor
			01	16-bit output from lower accumulator register, S
			10	16-bit output from lower 8x8

				multiplier, G
			11	Lower 16-bit output from 16x16 multiplier, H
BOTADDSUB_LOWERINPUT	C18,C17	Selects Input Z for the lower adder/subtractor	00	16-bit input from B input or associated pipeline register
			01	16-bit output from lower 8x8 multiplier, G
			10	Lower 16-bit output from 16x16 multiplier, H
			11	Sign extension input SIGNEXTIN. Duplicate 16 bits.
BOTADDSUB_UPPERINPUT	C19	Selects Input Y for the lower adder/subtractor	0	16-bit feedback from lower accumulator register, S
			1	16-bit input from D input or associated input register.
BOTADDSUB_CARRYSELECT	C21,C20	Carry/borrow input select to lower adder/subtractor	00	Constant 0
			01	Constant 1
			10	Cascade Carry/borrow input from ACCUMCI
			11	Carry/borrow input from CI
MODE_8x8	C22	Selects 8x8 Multiplier mode and 8x8 Low-Power Multiplier Blocking Option	0	No effect
			1	Selects 8x8 Multiplier mode. Holds the pipelining registers associate with a 16x16 multiplier in clock disable mode. Helps reduce the dynamic power consumption within the multiplier function. Used in conjunction with C6, C7 settings.
A_SIGNED	C23	Indicates whether multiplier input A is signed or unsigned. Applies regardless if input A is 16-or 32-bits wide.	0	Multiplier input A is unsigned.
			1	Multiplier input A is signed.
B_SIGNED	C24	Indicates whether multiplier input B is signed or unsigned. Applies regardless if input B is 16-or 32-bits wide.	0	Multiplier input B is unsigned.
			1	Multiplier input B is signed.

SB_MAC16 Configurations

The following SB_MAC16 functional blocks are supported in iCEcube2.

1. Multiplier.
2. Multiply and Accumulate (MAC).
3. Accumulator (ACC).
4. Add/Subtract (ADD/SUB).
5. Multiply and Add/Subtract (MULTADDSUB)

The valid configuration parameter settings for each functional block are listed below.

Note: Read the configuration settings from left to right while filling the SB_MAC16 instance parameter values.

Table 1 : Multiplier Configurations

8x8 Multiplier : 8x8 input multiplier with 16 bit output.

16x16 Multiplier: 16x16 input multiplier with 32 bit output.

SB_MAC16 Function/ Parameter Settings	B_SIGNED		A_SIGNED		MODE_8x8		BOTADDSUB_CARRYSELECT		BOTADDSUB_UPPERINPUT		BOTADDSUB_LOWERINPUT		BOTOUTPUT_SELECT		TOPADDSUB_CARRYSELECT		TOPADDSUB_UPPERINPUT		TOPADDSUB_LOWERINPUT		TOPOUTPUT_SELECT		PIPELINE_16x16_MULT_REG2		PIPELINE_16x16_MULT_REG1		BOT_8x8_MULT_REG		TOP_8x8_MULT_REG		D_REG		B_REG		A_REG		C_REG		
	CBIT[24]	CBIT[23]	CBIT[22]	CBIT[21]	CBIT[20]	CBIT[19]	CBIT[18]	CBIT[17]	CBIT[16]	CBIT[15]	CBIT[14]	CBIT[13]	CBIT[12]	CBIT[11]	CBIT[10]	CBIT[9]	CBIT[8]	CBIT[7]	CBIT[6]	CBIT[5]	CBIT[4]	CBIT[3]	CBIT[2]	CBIT[1]	CBIT[0]														
8x8 Multiplier																																							
mult_8x8_all_pipelined_unsigned	0	0	1	0	0	0	0	0	1	0	0	0	0	0	0	1	0	0	1	1	1	0	1	1	0														
mult_8x8_all_pipelined_signed	1	1	1	0	0	0	0	0	1	0	0	0	0	0	0	1	0	0	1	1	1	0	1	1	0														
mult_8x8_bypass_unsigned	0	0	1	0	0	0	0	0	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0														
mult_8x8_bypass_signed	1	1	1	0	0	0	0	0	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0														
16x16 Multiplier																																							
mult_16x16_all_pipelined unsigned	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	1	1	1	1	1	1	0	1	1	0														

mult_16x16_all_pipelined_signed	1	1	0	0	0	0	0	0	0	1	1	0	0	0	0	0	1	1	1	1	1	1	0	1	1	0
mult_16x16_intermediate_register_bypassed_unsigned	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0	1	1	1	0	0	0	0	1	1	0
mult_16x16_intermediate_register_bypassed_signed	1	1	0	0	0	0	0	0	1	1	0	0	0	0	0	0	1	1	1	0	0	0	0	1	1	0
mult_16x16_bypasses_unsigned	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0	1	1	0	0	0	0	0	0	0	0
mult_16x16_bypasses_signed	1	1	0	0	0	0	0	0	1	1	0	0	0	0	0	0	1	1	0	0	0	0	0	0	0	0

Table 2 : MAC Configurations

16 bit MAC: 8x8 input, 16 bit output MAC.

32 bit MAC: 16x16 input, 32 bit output MAC.

SB_MAC16 Function/ Parameter Settings	B_SIGNED		A_SIGNED		MODE_8x8		BOTADDSUB_CARRYSELECT		BOTADDSUB_UPPERINPUT		BOTADDSUB_LOWERINPUT		BOTOUTPUT_SELECT		TOPADDSUB_CARRYSELECT		TOPADDSUB_UPPERINPUT		TOPADDSUB_LOWERINPUT		TOPOUTPUT_SELECT		PIPELINE_16x16_MULT_REG2		PIPELINE_16x16_MULT_REG1		BOT_8x8_MULT_REG		TOP_8x8_MULT_REG		D_REG		B_REG		A_REG		C_REG	
	CBIT[24]	CBIT[23]	CBIT[22]	CBIT[21]	CBIT[20]	CBIT[19]	CBIT[18]	CBIT[17]	CBIT[16]	CBIT[15]	CBIT[14]	CBIT[13]	CBIT[12]	CBIT[11]	CBIT[10]	CBIT[9]	CBIT[8]	CBIT[7]	CBIT[6]	CBIT[5]	CBIT[4]	CBIT[3]	CBIT[2]	CBIT[1]	CBIT[0]													
16 bit MAC																																						
mac_16_all_pipelined_unsigned	0	0	1	0	0	0	0	1	0	1	0	0	0	0	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
mac_16_intermediate_register_bypassed_unsigned	0	0	1	0	0	0	0	1	0	1	0	0	0	0	1	0	1	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
mac_16_bypassed_unsigned	0	0	1	0	0	0	0	1	0	1	0	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
32 bit MAC																																						
mac_32_all_pipelined_unsigned	0	0	0	0	0	0	1	0	0	1	1	0	0	1	0	0	1	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

mac_32_all_pipeline_cascaded_unsigned	0	0	0	1	0	0	1	0	0	1	1	0	0	1	0	0	1	1	1	1	1	1	1	1	1
mac_32_all_pipeline_cin_unsigned	0	0	0	1	1	0	1	0	0	1	1	0	0	1	0	0	1	1	1	1	1	1	1	1	1
mac_32_intermediate_register_bypassed_unsigned	0	0	0	0	0	0	1	0	0	1	1	0	0	1	0	0	1	1	0	0	0	1	1	1	1
mac_32_intermediate_register_bypassed_cascaded_unsigned	0	0	0	1	0	0	1	0	0	1	1	0	0	1	0	0	1	1	0	0	0	1	1	1	1
mac_32_intermediate_register_bypassed_cin_unsigned	0	0	0	1	1	0	1	0	0	1	1	0	0	1	0	0	1	1	0	0	0	1	1	1	1
mac_32_bypassed_unsigned	0	0	0	0	0	0	1	0	0	1	1	0	0	1	0	0	1	0	0	0	0	0	0	0	0
mac_32_bypassed_cascaded_unsigned	0	0	0	1	0	0	1	0	0	1	1	0	0	1	0	0	1	0	0	0	0	0	0	0	0
mac_32_bypassed_cin_unsigned	0	0	0	1	1	0	1	0	0	1	1	0	0	1	0	0	1	0	0	0	0	0	0	0	0
mac_32_intermediate_register_bypassed_signed	1	1	0	0	0	0	1	0	0	1	1	0	0	1	0	0	1	1	0	0	0	1	1	1	1

Table 3 : ACCUMULATOR Configurations

16 bit ACCUMULATOR: 16 bit input, 16 bit output Accumulator.

32 bit ACCUMULATOR: 32 bit input, 32 bit output Accumulator.

16 bit	SB_MAC16 Function/ Parameter Settings
CBIT[24]	B_SIGNED
CBIT[23]	A_SIGNED
CBIT[22]	MODE_8x8
CBIT[21]	BOTADDSUB_CARRYSELECT
CBIT[20]	
CBIT[19]	BOTADDSUB_UPPERINPUT
CBIT[18]	
CBIT[17]	BOTADDSUB_LOWERINPUT
CBIT[16]	
CBIT[15]	BOTOUTPUT_SELECT
CBIT[14]	
CBIT[13]	TOPADDSUB_CARRYSELECT
CBIT[12]	
CBIT[11]	TOPADDSUB_UPPERINPUT
CBIT[10]	TOPADDSUB_LOWERINPUT
CBIT[9]	
CBIT[8]	TOPOUTPUT_SELECT
CBIT[7]	PIPELINE_16x16_MULT_REG2
CBIT[6]	PIPELINE_16x16_MULT_REG1
CBIT[5]	BOT_8x8_MULT_REG
CBIT[4]	TOP_8x8_MULT_REG
CBIT[3]	D_REG
CBIT[2]	B_REG
CBIT[1]	A_REG
CBIT[0]	C_REG